Nasr Kasrin (Dr.)

Product Thinker | Architect | Hippie Software Engineer

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n42r

Laid-back software architect, engineer, and product thinker with a critical eye and open mind. I shine on the intersection of engineering and product/people/cross-functional leadership *(see n42r.github.io)*

Projects

2023 – 2024 **Muze AI**, *Open Source Software*, Independent Project
AI/LLM music discovery and recommendation service *(see github.com/n42r/muze-ai)*
- **Roles:** Developer, architect.
- **Results:** First pure-LLM service that does not use the Spotify recommendation engine.

2022 – 2023 **Guestrrday**, *Open Source Software*, Independent Project
CLI tool for guessing year and label of songs *(see github.com/n42r/guestrrday)*
- **Roles:** Developer.
- **Results:** Successfully organized and tagged over 20,000 items.

2019 – 2023 **The Basin Network**, *Architectural Model/Invention*, Doctoral Project
Distributed data cataloging architectural model (cf. data mesh) *(see n42r.github.io/phd)*
- **Roles:** State-of-the-art technology analysis, standards development, researcher.
- **Results:** Invented a novel architecture, published 2 papers in international conferences.

2015 – 2019 **SIMUTOOL Data Lake**, *SaaS Platform*, EU-Funded Manufacturing Project
Data lake SaaS platform for an Industry 4.0 consortium *(see github.com/simutool)*
- **Roles:** Architect/API developer, team leader, software developer, product manager.
- **Results:** Used to manage & exchange data assets between 8 companies for 3 years.

2012 – 2014 **Greetings Studio**, *Social Network/Mobile App*, Startup (Employment)
Social network/iOS App for creating and sharing e-cards *(see n42r.github.io/gs)*
- **Roles:** Team leader, technical product manager, software architect.
- **Results:** Number of active users & rating steadily grew to 4 stars on the Apple Store.

2011 – 2013 **Tawla**, *Game/Mobile App*, Startup (Employment)
Backgammon board game App on the iOS mobile platform *(see n42r.github.io/tw)*
- **Roles:** R&D engineer: developed complex components (ex., AI players, randomization).
- **Results:** The rating of the App improved from 3 to 4+ stars citing the AI players.

2010 – 2011 **ArtSapiens**, *R&D Project (AI/Robotics)*, International RoboCup Competition
Software to compete in the international RoboCup AI Soccer competition.
- **Roles:** Co-founder of a team of 10+, head of R&D.
- **Results:** The software qualified and competed in the 2011 RoboCup finals.

2009 – 2010 **Cassie**, *R&D Project (AI/Robotics)*, Masters Project
Artificial Intelligence reasoning & control software for robot perception (Stack: LISP)
- **Results:** Discovered two novel methods, published 2 papers in top conferences.

Skills

Databases	MongoDB, Neo4j, SQL, Google Firebase Cloud, RDF
Development	Python3 (pydantic, mypy, pylint), Docker (multi-container), Linux CLI, git, UML
Past Skills	Lisp, C++, Java, PIC microcontroller programming (embedded)

Employment

2021 – 2023 **Freelance | Doctoral Student**
- Completed a PhD: A data cataloging architectural pattern *(see n42r.github.io/phd)*
- Several independent projects, explored emerging tools/technologies (ex., LLMs, APIs).

2015 – 2020 **Research Associate**, *EU Horizon2020 Project (University of Bamberg)*, Germany
- Managed a project to develop a data lake for a manufacturing consortium.
- Architect, team leader, product manager, developer *(see github.com/simutool)*.

2011 – 2014 **R&D Engineer | Architect | Team Leader**, *TayalT*, Cairo, Egypt
- Developed AI players for games, resulting in improved app rankings (C++, Java).
- Managed multiple products (social media apps and games) and reported to the CEO.
- Coordinated cross-functional teams of technical, marketing, UI/UX, and strategy.

Education

2023 **PhD. (Dr. rer. nat.)**, *Information Systems & Applied Computer Science, Otto-Friedrich-Universität Bamberg*, Bamberg, Germany

2010 **MSc.**, *Computer Science & Engineering, The German University in Cairo*, Egypt

2008 **BSc.**, *Computer Science & Engineering, The German University in Cairo*, Egypt

Publications

2021 Data-Sharing Markets for Integrating IoT Data Processing Functionalities. *CCF Transactions on Pervasive Computing and Interaction*.

2018 Semantic Data Management for Experimental Manufacturing Technologies. *Datenbank-Spektrum*.

2010 Focused Belief Revision as a Model of Fallible Relevance-Sensitive Perception. *Künstliche Intelligenz 2010: Advances in Artificial Intelligence*.

2010 High-Level Perception as Focused Belief Revision. *European Conference on AI (ECAI)*.

Languages

Arabic	Mother Tongue
English	Fluent
German	Good understanding, Basic Speaking (B1, Telc Certificate)